

Core Software Systems Global Universities

1. Identity & Access Layer

These control who can access what across the entire university.

1. Identity Management System
2. Single Sign-On (SSO) Platform
3. Multi-Factor Authentication System
4. Role & Permissions Management
5. Directory Services (User Database)

2. Student Academic Systems

These run the entire student lifecycle.

6. Student Information System (SIS)
7. Admissions & Application Management
8. Enrollment Management System
9. Course Registration System
10. Degree Planning / Academic Advising
11. Transcript & Academic Records System
12. Timetable / Class Scheduling System

3. Teaching & Learning Infrastructure

These power daily education delivery.

- 13. Learning Management System (LMS)
- 14. Assignment & Grading System
- 15. Online Assessment / Exam Platform
- 16. Lecture Capture System
- 17. Virtual Classroom / Video Teaching
- 18. Digital Content / Course Material Platform

4. Research Infrastructure

These manage grants, labs, and publications.

- 19. Research Information Management System
- 20. Grant Proposal Management
- 21. Research Funding Administration
- 22. Lab Management System
- 23. Research Data Management Platform
- 24. Institutional Repository / Publications System

5. Finance & Administration

These run the business side of the university.

- 25. Financial ERP System
- 26. Procurement / Vendor Management

- 27. Budget Planning System
- 28. Payroll System
- 29. Expense Management
- 30. Financial Reporting & Compliance

6. HR & Workforce Systems

Universities employ thousands of staff.

- 31. Human Resources Management System
- 32. Recruitment / Hiring System
- 33. Faculty Activity Reporting
- 34. Performance Management

7. Data & Intelligence Layer

These systems turn university data into decisions.

- 35. Data Warehouse / Institutional Data Platform
- 36. Business Intelligence / Analytics System
- 37. Institutional Reporting & Accreditation

8. Campus Infrastructure Systems

These run the physical campus.

- 38. Campus Facilities Management
- 39. IT Service Desk / Ticketing System

40. Digital Library Management System

Category	Systems
Identity	~5
Student lifecycle	~7
Teaching	~6
Research	~6
Finance/admin	~6
HR	~4
Data	~3
Campus operations	~3
Total ≈ 40 core systems	

Product	Replaces
University OS	Identity + SIS

Learning OS	LMS + Exams
Research OS	Grants + Labs
Finance OS	ERP
People OS	HR
Data OS	Analytics
Campus OS	Facilities
Library OS	Knowledge systems
Student Life OS	Housing + clubs
Collaboration OS	Communication
Automation OS	workflows
Integration OS	APIs

1. Identity & Access Layer

1. Identity Management System
2. Single Sign-On (SSO)

3. Multi-Factor Authentication (MFA)
4. Role & Permissions Management
5. Directory Services

2. Student & Academic Core

6. Student Information System (SIS)
7. Admissions & Application System
8. Enrollment Management System
9. Course Registration System
10. Degree Planning / Academic Advising
11. Academic Records & Transcript System
12. Timetable / Class Scheduling System
13. Curriculum Management System
14. Assessment & Accreditation System

3. Teaching & Learning Systems

15. Learning Management System (LMS)
 16. Virtual Classroom / Video Teaching
 17. Online Assessment / Examination System
 18. Lecture Capture System
- Assignment system - (inside LMS)
 - Grading system - (inside LMS)
 - Content system - (inside LMS)

4. Research Infrastructure

- 19. Research Information Management System
- 20. Grant Proposal & Funding System
- 21. Research Finance Tracking System
- 22. Lab Information Management System
- 23. Research Data Management Platform
- 24. Institutional Repository / Publications System
- 25. Research Compliance & Ethics System

5. Enterprise Finance & ERP

- 26. Enterprise ERP System

(Internally includes: finance, procurement, budgeting, expenses, reporting)

6. HR & Faculty Systems

- 27. Human Resource Management System
- 28. Recruitment / Hiring System
- 29. Performance Management System
- 30. Faculty Information System
 - teaching load
 - research output
 - tenure tracking

7. Data & Intelligence Layer

- 31. Institutional Data Warehouse
- 32. Business Intelligence / Analytics System
- 33. Institutional Reporting & Accreditation
- 34. Master Data Management (MDM) System

8. Campus & Operations

- 35. Facilities Management System
- 36. IT Service Desk / ITSM System
- 37. Library Management System
- 38. Room Scheduling System (often separate from SIS)
- 39. Campus Card / Identity System
- 40. Physical Security & Surveillance System

9. Integration Layer

- 41. Integration Platform (API / Middleware / ESB)

10. Infrastructure Layer

- 42. Cloud & Infrastructure Management System
- 43. Network Management System

11. Cybersecurity Layer

- 44. Security Operations / SIEM System

Extras

- 45. Identity Governance & Administration (IGA)
- 46. API Management Platform
- 47. Workflow / BPM Automation Platform
- 48. Enterprise Content & Records Management System
- 49. Backup & Disaster Recovery System
- 50. Observability / Monitoring Platform
- 51. Endpoint / Device Management System